

SAFETY MANUAL

FLOATING AND FULL TRUNNION

BALL VALVES

PEKOS GROUP

INDEX

1. INTRODUCTION

- 1.1 Terms and Abbreviations
- 1.2 Acronyms
- 1.3 Product Support
- 1.4 Related Literature
- 1.5 Reference Standards

2. DEVICE DESCRIPTION

3. DESIGNING A SIF USING A PEKOS FLOATING AND FULL TRUNNION BALL VALVES

- 3.1 Safety Function
- 3.2 Environmental limits
- 3.3 Application limits
- 3.4 Design Verification
- 3.5 SIL Capability
 - 3.5.1 *Systematic Integrity*
 - 3.5.2 *Random Integrity*
 - 3.5.3 *Safety Parameters*
- 3.6 Connection of the Floating and Full Trunnion ball valves to the SIS Logic-solver
- 3.7 General Requirements

4 INSTALLATION AND COMMISSIONING

- 4.1 Installation

5. OPERATION AND MAINTENANCE

- 5.1 Proof test without automatic testing
- 5.2 Proof test with automatic partial valve stroke testing
- 5.3 Repair and replacement
- 5.4 Warnings of operation and maintenance
- 5.5 Useful Life
- 5.6 PEKOS Notification

1. INTRODUCTION

This Safety Manual provides information necessary to design, install, verify and maintain a Safety Instrumented Function (SIF) utilizing the PEKOS Floating and Full Trunnion ball Valves. This manual provides necessary requirements for meeting the IEC 61508 or IEC 61511 functional safety standards.

1.6 Terms and Abbreviations

Safety	Freedom from unacceptable risk of harm
Functional Safety	The ability of a system to carry out the actions necessary to achieve or to maintain a defined safe state for the equipment / machinery / plant / apparatus under control of the system
Basic Safety	The equipment must be designed and manufactured such that it protects against risk of damage to persons by electrical shock and other hazards and against resulting fire and explosion. The protection must be effective under all conditions of the nominal operation and under single fault condition
Safety Assessment	The investigation to arrive at a judgment - based on evidence - of the safety achieved by safety-related systems
Fail-Safe State	State where solenoid valve is de-energized and spring is extended.
Fail Safe	Failure that causes the valve to go to the defined fail-safe state without a demand from the process.
Fail Dangerous	Failure that does not respond to a demand from the process (i.e. being unable to go to the defined fail-safe state).
Fail Dangerous Undetected	Failure that is dangerous and that is not being diagnosed by automatic stroke testing.
Fail Dangerous Detected	Failure that is dangerous but is detected by automatic stroke testing.
Fail Annunciation Undetected	Failure that does not cause a false trip or prevent the safety function but does cause loss of an automatic diagnostic and is not detected by another diagnostic.
Fail Annunciation Detected	Failure that does not cause a false trip or prevent the safety function but does cause loss of an automatic diagnostic or false diagnostic indication.
Fail No Effect	Failure of a component that is part of the safety function but that has no effect on the safety function.
Low demand mode	Mode, where the frequency of demands for operation made on a safety-related system is no greater than twice the proof test frequency.

1.7 Acronyms

FMEDA	Failure Modes, Effects and Diagnostic Analysis
HFT	Hardware Fault Tolerance
MOC	Management of Change. These are specific procedures often done when performing any work activities in compliance with government regulatory authorities.
PFD _{AVG}	Average Probability of Failure on Demand
SFF	Safe Failure Fraction, the fraction of the overall failure rate of a device that results in either a safe fault or a diagnosed unsafe fault.
SIF	Safety Instrumented Function, a set of equipment intended to reduce the risk due to a specific hazard (A safety loop).
SIL	Safety Integrity Level, discrete level (one out of a possible four) for specifying the safety integrity requirements of the safety functions to be allocated to the E/E/PE safety-related systems where Safety Integrity Level 4 has the highest level of safety integrity and Safety Integrity Level 1 has the lowest.
SIS	Safety Instrumented System – Implementation of one or more Safety Instrumented Functions. A SIS is composed of any combination of sensor(s), logic solver(s), and final element(s).

1.8 Product Support

Product support can be obtained from:

PEKOS GROUP

CONTACT INFORMATION

C/ Rec del Molinar, 9 – P.I. El Circuit

E-08160 Montmeló (Barcelona)

Tel (+34) 93 579 93 70

Fax: (+34) 93 579 92 44

www.pekos.es

1.9 Related Literature

Hardware Documents:

- Manual Instruction For Storage, Installation, Operation And Maintenance Of Pekos Ball Valve

Guidelines/References:

- Safety Integrity Level Selection – Systematic Methods Including Layer of Protection Analysis, ISBN 1-55617-777-1, ISA
- Control System Safety Evaluation and Reliability, 2nd Edition, ISBN 1-55617-638-8, ISA
- Safety Instrumented Systems Verification, Practical Probabilistic Calculations, ISBN 1-55617-909-9, ISA

1.10 Reference Standards

Functional Safety

- IEC 61508: 2010 Functional safety of electrical/electronic/ programmable electronic safety-related systems
- ANSI/ISA 84.00.01-2004 (IEC 61511 Mod.) Functional Safety – Safety Instrumented Systems for the Process Industry Sector

2. DEVICE DESCRIPTION

The Floating and Full Trunnion ball valves are $\frac{1}{4}$ turn ball valves used to control process fluids.

The valves are available from 0,5" to 30" and are designed to meet international standards for pressure and temperature ratings, shell thickness, and bore diameters. The Floating and Full Trunnion ball valves provides ISO 5211 mounting for simple actuator mounting. The Floating and Full Trunnion ball valves include double body-gasket sealing and multiple stem seals.

3. DESIGNING A SIF USING A PEKOS FLOATING AND FULL TRUNNION BALL VALVES

3.1 Safety Function

When de-energized, the actuator moves the Floating and Full Trunnion ball valves to their failsafe position. Depending on the version specified Fail – Closed or Fail - Open, the Floating and Full Trunnion ball valves will rotate the valve plug to close off the flow path through the valve body or open the flow path through the valve body.

The Floating and Full Trunnion ball valves is intended to be part of final element subsystem as defined per IEC 61508 and the achieved SIL level of the designed function must be verified by the designer.

3.2 Environmental limits

The designer of a SIF must check that the product is rated for use within the expected environmental limits. Refer to Manual Instruction for Storage, Installation, Operation and Maintenance of Pekos Ball Valve

* For valves placed in plants in particularly humid areas, in the nearby of the sea or other hostile places the material of the valve will have to be taken in consideration together with external protective painting as far as external corrosion is concerned.

3.3 Application limits

The materials of construction of a Floating and Full Trunnion ball valves are specified in the 3.1 certificate that Pekos Group sends to the client. It is especially important that the designer check for material compatibility considering on-site chemical contaminants and air supply conditions. If the Floating and Full Trunnion ball valves are used outside of the application limits or with incompatible materials, the reliability data provided becomes invalid. For the storage of some types of materials, it can also be consult the Manual Instruction for Storage, Installation, Operation and Maintenance of Pekos Ball Valve.

3.4 Design Verification

A detailed Failure Mode, Effects, and Diagnostics Analysis (FMEDA) report is available from Pekos. This report details all failure rates and failure modes as well as the expected lifetime.

The achieved Safety Integrity Level (SIL) of an entire Safety Instrumented Function (SIF) design must be verified by the designer via a calculation of PFD_{AVG} considering architecture, proof test interval, proof test effectiveness, any automatic diagnostics, average repair time and the specific failure rates of all products included in the SIF. Each subsystem must be checked to assure compliance with minimum hardware fault tolerance (HFT) requirements. The *exida* exSILentia® tool is recommended for this purpose as it contains accurate models for the Floating and Full Trunnion ball valves and their failure rates.

When using Floating and Full Trunnion ball valves in a redundant configuration, a common cause factor of at least 5% should be included in safety integrity calculations.

The failure rate data listed the FMEDA report is only valid for the useful life time of a Floating and Full Trunnion ball valves. The failure rates will increase sometime after this time period. Reliability calculations based on the data listed in the FMEDA report for mission times beyond the lifetime may yield results that are too optimistic,

	<i>SAFETY MANUAL</i> <i>Floating and Full Trunnion ball Valves</i>	<i>DC 77-02-04</i> <i>Rev. 3</i>
--	--	--

i.e. the calculated Safety Integrity Level will not be achieved.

3.5 SIL Capability

3.5.1 Systematic Integrity

The product has met PEKOS design process requirements of Safety Integrity Level (SIL) 3. These are intended to achieve sufficient integrity against systematic errors of design by the manufacturer. A Safety Instrumented Function (SIF) designed with this product must not be used at a SIL level higher than the statement without “prior use” justification by end user or diverse technology redundancy in the design.

3.5.2 Random Integrity

When the final element assembly consists of many components (Floating and Full Trunnion ball valves, actuator, solenoid, quick exhaust valve, etc.) the SIL must be verified for the entire assembly using failure rates from all components. This analysis must account for any hardware fault tolerance and architecture constraints.

3.5.3 Safety Parameters

For detailed failure rate information refer to the Failure Modes, Effects and Diagnostic Analysis Report for the Floating and Full Trunnion ball valves.

3.6 Connection of the Floating and Full Trunnion ball valves to the SIS Logic-solver

The Floating and Full Trunnion ball valves are connected to the safety rated logic solver which is actively performing the safety function as well as automatic diagnostics designed to diagnose potentially dangerous failures within the Floating and Full Trunnion ball valves (i.e. partial valve stroke test).

3.7 General Requirements

The system’s response time shall be less than process safety time. All SIS components including the Floating and Full Trunnion ball valves must be operational before process start-up.

User shall verify that the Floating and Full Trunnion ball valves are suitable for use in safety applications by confirming the Floating and Full Trunnion ball valves nameplates are properly marked.

Personnel performing maintenance and testing on the Floating and Full Trunnion ball valves shall be competent to do so.

Results from the proof tests shall be recorded and reviewed periodically.

The useful life of the Floating and Full Trunnion ball valves is discussed in the Failure Modes, Effects and Diagnostic Analysis Report for the Floating and Full Trunnion ball valves.

4 INSTALLATION AND COMMISSIONING

4.1 Installation

The Floating and Full Trunnion ball valves valve must be installed per standard practices outlined in Manual Instruction For Storage, Installation, Operation And Maintenance Of Pekos Ball Valve.

The environment must be checked to verify that environmental conditions do not exceed the ratings.

The Floating and Full Trunnion ball valves must be accessible for physical inspection.

5. OPERATION AND MAINTENANCE

5.1 Proof test without automatic testing

The objective of proof testing is to detect failures within a Pekos valve that are not detected by any automatic diagnostics of the system. Of main concern are undetected failures that prevent the safety instrumented function from performing its intended function.

The frequency of proof testing, or the proof test interval, is to be determined in reliability calculations for the safety instrumented functions for which an Pekos valve is applied. The proof tests must be performed more frequently than or as frequently as specified in the calculation in order to maintain the required safety integrity of the safety instrumented function.

The following proof test is recommended. The results of the proof test should be recorded and any failures that are detected and that compromise functional safety should be reported to Pekos. The suggested proof test consists of a full stroke of the Floating and Full Trunnion ball valves.

Step	Action
1	Bypass the safety function and take appropriate action to avoid a false trip.
2	Send a signal to the final element configuration to perform a full stroke and verify that this is achieved.
3	Inspect the Floating and Full Trunnion ball valves for any visible damage or contamination.
4	Record any failures in your company's SIF inspection database.
5	Remove the bypass and otherwise restore normal operation.

Table1: Recommended Proof Test

This test will detect >47% of possible DU failures in the Floating and Full Trunnion ball valves for the Full Stroke options. (For the Tight Shutoff and Open to Trip options the proof test coverage is >14%, and > 92% respectively). If an additional leakage test is performed, the DC (Diagnostic Coverage) will increase substantially.

The person(s) performing the proof test of a Floating and Full Trunnion ball valves should be trained in SIS operations, including bypass procedures, valve maintenance and company Management of Change procedures. No special tools are required.

5.2 Proof test with automatic partial valve stroke testing

An automatic partial valve stroke testing scheme that performs a full stroke of the isolation valves in the Floating and Full Trunnion ball valves and measures valve movement timing will detect most potentially dangerous failure modes. It is recommended that a physical inspection (Step 2 from Table 1) be performed on a periodic basis with the time interval determined by plant conditions. A maximum inspection interval of five years is recommended.

5.3 Repair and replacement

Repair procedures in the Manual Instruction For Storage, Installation, Operation And Maintenance Of Pekos Ball Valve.

5.4 Warnings of operation and maintenance

- a) The end user must have suitable safety and control devices to avoid over pressure and over temperature. The end user must check that the surface temperatures are lower than the flashpoint temperature of the surrounding environment, or inner surface dust.
- b) The use of the valves in a semi open position for a long time can damage the body seat material.
- c) The end user will have to foreseen external factors like: freezing, dynamic effects due to the line vibration, exceptional phenomena like wind, earthquakes, flooding, etc.
- d) Valves without heating jackets will not be used for medium that, due to the temperature, can harden or crystallize.
- e) Valves without stem extension cannot be used on insulated lines.
- f) Valves without PEKOS locking system cannot be used where it can be dangerous the accidental opening or closing of the valve.
- g) Floating valves without supplementary hole cannot be used in case of instable fluids which, if trapped in the body cavity (valves in closed position) can be subject to the risk of increasing of pressure due to the increasing of temperature.
- h) Valves without cavity filler and special cleaning / polishing should not be used for hygienic service.
- i) Valves without fugitive emission extensions cannot be used for toxic or dangerous medium with the risk of dispersion in the environment.
- j) For fluids like oxygen, hydrogen or chlorine where the contact with oil or grease can create explosions valves that have not been properly cleaned, degreased and sealed in suitable boxes cannot be used. Avoid any kind of contamination till the moment of use. The same precautions must apply to valves for cryogenic services.
- k) The maintenance has to be made by person authorized by PEKOS, using original repair kits.
- l) PEKOS recommends inspecting the valves at least every five (5) years. These inspection intervals could be affected by the process service (fluid, temperature, pressure, and cycles), and environmental conditions. For further information, see applicable manual instruction for storage, installation, operation and maintenance (IOM).

5.5 Useful Life

The useful life of the Floating and Full Trunnion ball valves is **10 to 15 years**.

5.5 PEKOS Notification

Any failures that are detected and that compromise functional safety should be reported to **PEKOS GROUP**:

C/ Rec del Molinar, 9 – P.I. El Circuit

E-08160 Montmeló (Barcelona)

Tel (+34) 93 579 93 70

Fax: (+34) 93 579 92 44

www.pekos.es